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PyRayHF: An Open Source Ray-Tracing Forward Operator for Assimilating Oblique Ionograms

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Key Points:

- `PyRayHF` is an open-source, Python-based framework for HF ionospheric forward modeling
- It provides magneto-ionic vertical forward operators and multiple 2D ray tracing formulations for both stratified and structured plasmas
- `PyRayHF` includes a parameterized minimization scheme that enables assimilation of midpoint-converted oblique HF observations

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Abstract Oblique high-frequency (HF) ionospheric measurements contain valuable information about electron density structure, but remain difficult to assimilate operationally due to the computational cost of ray tracing. `PyRayHF` is an open-source, fully Python-based framework that addresses such difficulties through the implementation of efficient forward operators for HF propagation and parameter retrieval tailored to data-assimilation (DA) applications. `PyRayHF` implements magneto-ionic vertical tracing and a hierarchy of two-dimensional oblique ray tracing methods, which are verified against one another for internal consistency and validated through comparative analysis with a full three-dimensional Jones-Stephenson ray tracer. Validation results show that all solvers reproduce key observables, including group path and apex location, with typical errors below 10% when compared to the reference solution. For group path, the primary observable in oblique ionogram interpretation, simple stratified Snell's-law formulations perform comparably to gradient-based solvers while being several orders of magnitude faster, making them suitable for DA workflows. To enable efficient assimilation, `PyRayHF` combines a fast vertical forward operator with a midpoint-based oblique-to-vertical transformation. For transmitter–receiver separations of 1,000–2,000 km, midpoint-derived $foF2$ errors remain below 10%. `PyRayHF` further provides a parameter-minimization framework that retrieves F2-layer parameters directly from midpoint-converted observations, producing anchor-point measurements suitable for direct assimilation into parametrized systems such as ANCHOR.

Plain Language Summary High-frequency (HF) radio signals may reflect off of the ionosphere, a region of Earth's upper atmosphere containing charged particles. Measurements of these reflected signals can provide useful information about conditions in the ionosphere. However, such measurements are difficult to interpret and include in space weather models because doing so typically requires the use of computationally expensive ray-tracing simulations. `PyRayHF` is a lightweight, open-source, Python package that provides (a) accessible ray-tracing tools for modeling HF signal propagation through the ionosphere and (b) a practical method of extracting useful ionospheric data from measured HF signals. In this paper, the underlying physics and core design principles of this software are presented. The ray-tracing methods within `PyRayHF` are verified against one another to ensure internal consistency and these methods are further tested against an existing, state-of-the-art, 3-dimensional ray tracer to assess accuracy. `PyRayHF`'s built-in methods for extracting ionosphere data from HF signals are demonstrated and techniques for incorporating these measurements into existing models are also discussed.

1. Introduction

The ionosphere is the region of Earth's upper atmosphere, extending from roughly 80 to 1,000 km altitude, where solar radiation ionizes neutral particles to form a partially conducting plasma. Variations in its electron density structure strongly influence the propagation of high-frequency (HF, 3–30 MHz) radio waves, enabling over-the-horizon radar observations and long-range communications through the refraction or reflection of signals back to Earth. Understanding and modeling the ionosphere and its interactions with HF signals is therefore essential across sectors. Commercial operators rely on accurate ionospheric forecasts to mitigate disruptions in global navigation satellite system (GNSS) positioning and satellite communications. Academic researchers need precise measurement geolocation to support investigation into space-weather processes and magnetosphere–ionosphere coupling. Additionally, the defense community depends on accurate models for reliable HF communication, radar performance, and navigation resilience.

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Radio propagation within the ionosphere is highly sensitive to solar and geomagnetic variations in electron density. This sensitivity makes three-dimensional reconstructions of the plasma structure crucial for predicting signal behavior. In recent years, ionospheric modeling has increasingly emphasized real-time specification through data assimilation (DA), in which diverse observations are merged with physics-based models. Examples of prominent DA systems include the Ionospheric Data Assimilation Four-Dimensional model (IDA4D) (Bust & Crowley, 2007), the Global Assimilation of Ionospheric Measurements (GAIM) (Schunk et al., 2004), the MAGIC United States Total-Electron Content (US-TEC) framework (Fuller-Rowell et al., 2006), and the Global Ionospheric Specification (GIS) model (Lin et al., 2020). More recently, the U.S. Naval Research Laboratory has developed ANCHOR (V. V. Forsythe, McDonald, et al., 2024), a Kalman-filter-based DA system intended for operational space-weather nowcasting.

Most existing DA systems assimilate vertical soundings, GNSS-TEC, and radio-occultation data because these observables are relatively linear, abundant, and easy to relate to electron density fields. In contrast, direct assimilation of oblique HF measurements remains an emerging frontier. The challenge arises from the inherently nonlinear relationship between HF ray paths and the three-dimensional electron density distribution. This nonlinearity requires embedding or approximating full ray-tracing operators and their sensitivity functions within the DA framework.

One approach to assimilating oblique HF ray paths has been implemented within the Global Positioning System Ionospheric Inversion (GPSII) system (Fridman et al., 2006, 2009, 2012). GPSII introduced a Ray-Path Response Operator to linearize the relationship between small electron-density perturbations and oblique path observables. Fridman et al. (2016) demonstrated that assimilating multi-channel HF link data within this framework significantly improves ionospheric reconstructions. In a complementary effort, Martin et al. (2017) explored assimilation of forward-modeled oblique ionosonde (FOI) profiles into the Electron Density Assimilative Model (EDAM). Using synthetic experiments, they showed that FOI assimilation improves bottomside electron density estimates and better reproduces withheld oblique traces. They also found that sensitivity above the F2 peak remains limited. Their study represents one of the earliest demonstrations that oblique ionosonde data can be incorporated into a three-dimensional DA system.

An alternative approach to incorporating oblique sounding observations into ionospheric modeling is to derive equivalent vertical ionogram representations from the measured oblique ionograms. Heitmann and Gardiner-Garden (2019) developed a robust feature extraction and parameterized fitting algorithm for both oblique and vertical incidence ionograms, demonstrating how key ionogram characteristics can be automatically identified and represented in a form suitable for subsequent modeling and analysis. Such methods provide a pathway for incorporating oblique sounding information into ionospheric reconstruction and assimilation frameworks while avoiding the need to directly assimilate the raw ionogram measurements.

Despite these advances, the direct assimilation of oblique ionosphere sounding data is not commonplace. No publicly known operational DA frameworks perform comprehensive oblique HF assimilation with full ray-tracing sensitivity. The main obstacle is the absence of computationally efficient, openly accessible forward operators that map background electron density to key HF observables, such as group delay, ground range, and oblique group path, for specified frequencies and elevation angles.

Several ray-tracing frameworks have been developed for ionospheric propagation studies, including PHaRLAP (Cervera & Harris, 2014), IONORT (Pietrella et al., 2023), and IONOLAB-RAY (Erdem & Arikan, 2017), and various descendants of the Jones-Stephenson model (Jones & Stephenson, 1975a), such as (Bristow & Greenwald, 1995). More recent developments include mixed-language and C++ frameworks such as AU-Ray (Hirvonen et al., 2025) and FARR (Green et al., 2025), which offer magneto-ionic, three-dimensional capabilities. Collectively, these tools form a scattered but functional ecosystem for HF and VHF propagation modeling. This ecosystem remains dominated by compiled languages for numerical efficiency and compatibility with legacy infrastructures.

Ray tracing tools have been incorporated into ionospheric reconstruction and assimilation studies. For example, Erdem et al. (2015) presented a model-based ray-tracing approach coupled with GIM-TEC-based ionospheric reconstruction. In that work, ray tracing was applied to a previously reconstructed ionosphere and served as a propagation modeling tool. The present study, however, addresses a different problem: the development of a ray-tracing forward operator that allows oblique ionogram observations themselves to be assimilated and used to

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constrain the ionospheric state. The primary challenge considered here is not high fidelity propagation through a reconstructed ionosphere, but the computationally efficient evaluation of large numbers of forward-model simulations required during iterative local DA processes.

Despite the variety of available tools, few of these are suitable for DA applications. A conventional ray tracer launches rays over multiple elevation angles and performs path homing. When all 3D magneto-ionic effects are considered, this typically requires 10–30 s to generate a single oblique ionogram. Even without incorporating a full DA framework, consider a simple minimization problem: estimating the peak electron density *Nm*F2 and its corresponding height *hm*F2 between a transmitter and receiver to best match an observed ionogram. Such an optimization would necessitate executing the ray tracer hundreds of times for successive electron-density updates, potentially requiring hours of computational time. In the context of DA, where this process would need to be repeated for *all* available oblique ionograms, the allowable runtime for an operational assimilation cycle would be greatly exceeded.

At the U.S. Naval Research Laboratory (NRL), several Python-based ionospheric tools have become integral components of both operational DA systems and the broader modeling infrastructure. The PyIRI model (V. V. Forsythe, Bilitza, et al., 2024; V. V. Forsythe, Servan-Schreiber, et al., 2025) is widely used for scientific studies and serves as a background specification source within operational workflows. The ANCHOR system (V. V. Forsythe, McDonald, et al., 2024) is likewise implemented entirely in Python, encompassing data ingestion, aggregation, and the DA engine itself. Complementary validation software, PyVALION (V. V. Forsythe, Burrell, & Evans, 2025), is also written in Python. These tools share common libraries and are designed for interoperability within a unified NRL ecosystem. Consequently, development of a Python-based HF ray tracer represents a natural extension of this architecture. Such a tool enables direct assimilation of oblique HF observations into ANCHOR, closing a critical capability gap in current ionospheric DA efforts.

This paper has four main objectives. First, it serves an educational role by illustrating the governing physics of HF signal propagation in the ionosphere (Section 2). Second, it introduces `PyRayHF` (V. Forsythe, Burrell, Valentine, & Hughes, 2025), a software framework that provides a suite of HF forward operators for vertical propagation (Section 3) and two-dimensional oblique ray tracing (Section 4). Third, it verifies the developed components of `PyRayHF` by ensuring their internal consistency (Section 5) and comparing HF forward operator outputs to those of a full 3D magneto-ionic ray tracer (Section 6). Finally, it examines HF propagation from a DA perspective and outlines strategies for integrating parametrized HF operators into assimilation systems such as ANCHOR (Sections 7–9).

2. The Physics of HF Ray Propagation

Wave propagation in the ionosphere can be characterized by two refractive indices. The phase refractive index μ determines how the phase of an electromagnetic wave advances through plasma and is defined as the ratio between the speed of light c and the phase velocity of the wave v_ϕ,

$$\mu = \frac{c}{v_\phi} \tag{1}$$

The group refractive index μ' governs the propagation of wave energy or a modulation envelope. It is related to the phase index through the frequency dependence

$$\mu' = \mu + f\,\frac{\partial \mu}{\partial f}, \tag{2}$$

where f is the transmission frequency of the wave and the prime indicates that μ' is derived from μ via dispersion. The corresponding group velocity is $v_g = c/\mu'$. This velocity controls the travel time of the ionosonde pulse and therefore the measured virtual height.

For a cold, magnetized electron plasma with fixed ions and a monochromatic wave of frequency f, the geomagnetic field makes the medium anisotropic. Propagation must therefore be described using magneto-ionic theory. Neglecting collisions, the phase refractive index is given by the Appleton–Hartree equation (Budden, 1985; Davies, 1990)

$$\mu^2 = 1 - \frac{X(1-X)}{1 - X - \frac{1}{2}Y^2\sin^2\theta \pm \sqrt{\frac{1}{4}Y^4\sin^4\theta + (1-X)^2Y^2\cos^2\theta}}. \quad (3)$$

We define

$$X = \frac{f_p^2}{f^2}, \qquad Y = \frac{f_c}{f}, \quad (4)$$

where the plasma frequency f_p is computed in terms of the elementary charge e, the mass of an electron m_e, the permittivity of free space ε_0, and the electron density N_e in electrons/m^3

$$f_p = \frac{1}{2\pi}\sqrt{\frac{N_e e^2}{\varepsilon_0 m_e}} = 8.97866275\sqrt{N_e}. \quad (5)$$

f_c is the electron gyrofrequency, obtained from the expression

$$f_c = \frac{eB}{2\pi m_e} = \left(2.799249 \times 10^{10}\right)B, \quad (6)$$

where the constant terms have been combined and the magnetic field B is provided in units of tesla. The angle θ is defined between the wave vector and the magnetic field.

A wave traveling vertically through a stratified ionosphere will reflect when the phase refractive index reaches zero. The conditions for reflection may therefore be obtained by evaluating the roots of the Appleton-Hartree equation. Substituting $\mu = 0$ into Equation 3 and rearranging terms yields

$$Y^2\sin^2\theta - 2(1-X)^2 = \pm\sqrt{Y^4\sin^4\theta + 4(1-X)^2Y^2\cos^2\theta}. \quad (7)$$

The "±" operator in Equations 3 and 7 suggests the existence of two characteristic propagation modes, each with its own reflection condition. The first of these, corresponding to the "+" solution, is the ordinary mode, or O-mode. In this case, Equation 7 is satisfied when $X = 1$ for all $\theta \neq 0°$. According to the definition of X provided in Equation 4, this requires $f = f_p$. Thus, for O-mode propagation, reflection occurs at an altitude where the plasma frequency matches the transmission frequency and the maximum returned ionosonde frequency $f_{\max}$ corresponds to the maximum, or *critical*, value of f_p,

$$f_{\max} = f_{p_{\text{crit}}}. \quad (8)$$

The "−" case of Equations 3 and 7 corresponds to the extraordinary mode or X mode. The X mode refractive index has roots at $X = 1 \pm Y$ for all propagation directions. For typical ionospheric densities and HF transmission frequencies $>\sim 3$ MHz, $Y < 1$ is satisfied. Consequently, the physically relevant cutoff condition is $X = 1 - Y$, as this is the first root encountered with increasing plasma density. Unlike the O-mode, this cutoff condition depends directly on the magnetic field strength due to the presence of the Y term. Substituting the definitions of X and Y into $X = 1 - Y$ yields,

$$\frac{f_p^2}{f^2} = 1 - \frac{f_c}{f}. \quad (9)$$

This expression may be rearranged to obtain the maximum returned X-mode ionosonde frequency corresponding to the critical plasma frequency,

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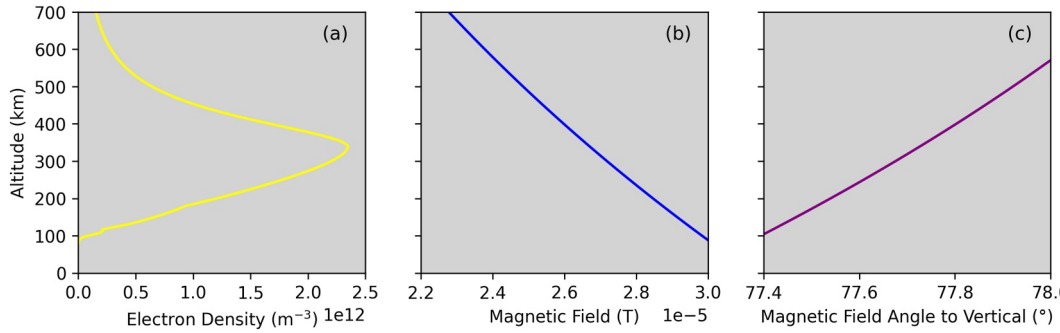


Figure 1. Example input arrays for vertical ray tracing, generated with the PyIRI model for conditions at 150°W, 4.5°N (F10.7 = 204 SFU) on 1 September 2025, 00:00 UT.

$$f_{\max} = \frac{f_c + \sqrt{f_c^2 + 4f_{p_{\text{crit}}}^2}}{2}. \tag{10}$$

3. Vertical Forward Operator

An ionosonde transmits short high-frequency pulses vertically into the ionosphere and records the time delay of echoes returned at different reflection frequencies. The recorded quantity is the virtual height $h_v(f)$, which represents the apparent altitude of reflection if the wave propagated at the speed of light in free space. In reality, the signal is gradually refracted by the ionized medium, and the true reflection point occurs at a lower physical altitude h_t. The difference between these heights arises from plasma dispersion, which reduces the group velocity of the wave relative to c.

The apparent virtual height observed by an ionosonde is obtained by integrating the group refractive index along the vertical path,

$$h_v(f) = \int_0^{h_t} \mu'(h', f)\, \mathrm{d}h', \tag{11}$$

where h_t is the true reflection height. For the O-mode, h_t is located where $X = 1$. For the X-mode, h_t is located where $X + Y = 1$. This integral forms the basis of the `PyRayHF` vertical forward operator. Given an electron density profile $N_e(h)$ and the local magnetic field $B(h)$, μ' is evaluated and the operator computes the frequency-dependent virtual height $h_v(f)$ for both O- and X-mode traces.

Figure 1 illustrates example input arrays for the `PyRayHF` vertical operator. Panels (a), (b), and (c) show the electron density profile, the magnetic field magnitude, and the magnetic field inclination angle Ψ relative to the local vertical, respectively. These data were modeled using PyIRI v0.1.5 (V. V. Forsythe, Servan-Schreiber, et al., 2025) at 150°W, 0°N on 1 September 2025, 00:00 UT, using a solar flux index of F10.7 = 204 SFU.

In `PyRayHF`, the critical height is determined by linearly interpolating the altitude grid to locate where the cutoff conditions are satisfied. Accurate integration immediately below this reflection height is essential because the group refractive index μ' varies rapidly in this region. To achieve high resolution near the cutoff while maintaining computational efficiency elsewhere, a stretched vertical grid is employed. The following expression defines the altitude of each grid point z_i

$$z_i = a_0 + (a_{\text{crit}} - a_0)\, S(\xi_i), \tag{12}$$

where a_0 is the minimum altitude, a_{crit} is the frequency-dependent critical height, and $S(\xi_i)$ is a smooth monotonic mapping function that compresses the grid spacing near the upper boundary. The mapping function is depicted in Figure 2a and is defined by

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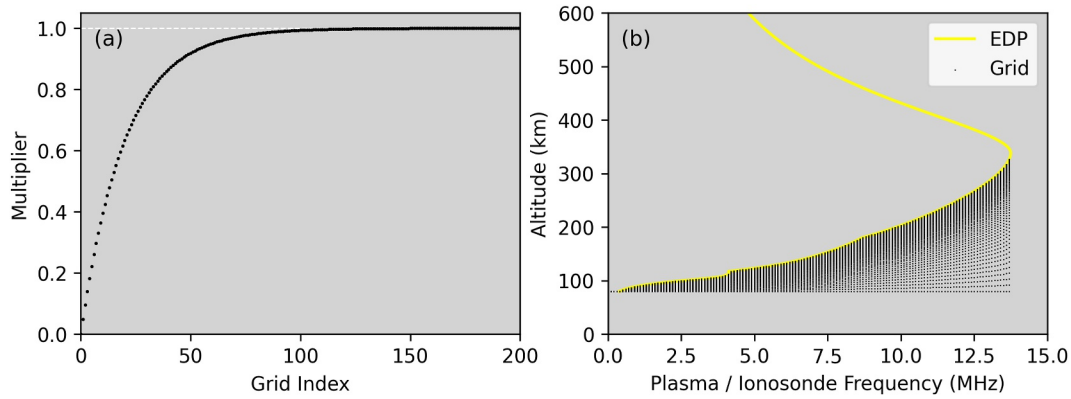


Figure 2. (a) Smooth monotonic mapping function $S(\xi_i)$. (b) Distribution of grid points relative to Figure 1 electron density profile (EDP). The EDP density is represented in terms of plasma frequency according to Equation 5.

$$S(\xi) = 1 - \frac{\exp[\alpha(1-\xi)] - 1}{\exp(\alpha) - 1}, \tag{13}$$

with $\xi \in [0, 1]$ and $\alpha = 10$ controlling the degree of stretching. As demonstrated in Figure 2b, this construction concentrates grid points near the reflection height where precise numerical integration of μ' is most critical and allows for coarser resolution at lower altitudes.

To further improve computational efficiency, all input profiles are transformed into matrices of shape $[N_f, N_g]$, where N_f is the number of ionosonde frequencies and N_g is the number of points in the stretched vertical grid. This transformation is performed using one-dimensional interpolation, resampling each physical quantity onto the non-uniform grid. Figure 3 shows the previously obtained altitude, electron density, magnetic-field magnitude, and magnetic-field inclination angle expressed in this matrix form. The physical information is unchanged from Figure 1. The only difference is the stretched vertical coordinate, which enables more efficient vectorized computation of $h_v(f)$ within Python.

Figure 4 presents intermediate quantities computed on the stretched grid. Panels (a) and (b) show the dimensionless parameters X and Y, which represent the relative contributions of plasma density and magnetic field, respectively. The parameter X increases from zero at the bottom of the ionosphere to nearly unity in the reflection region. Because of the grid stretching, values of $X \approx 1$ occur around index 75 of 200, where the vertical resolution is highest. In contrast, Y exhibits little vertical variation. It depends primarily on magnetic-field strength and transmitted frequency and therefore varies mainly along the frequency dimension, reaching values of approximately 0.85 under equatorial conditions.

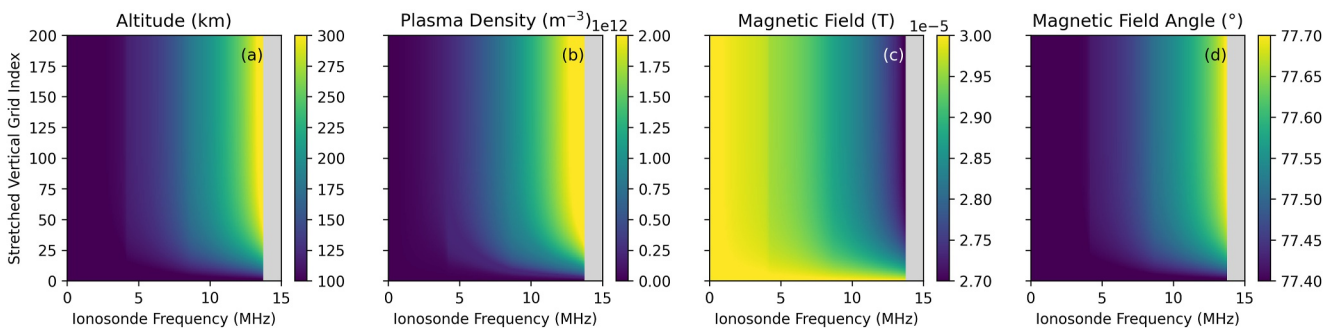


Figure 3. Input arrays interpolated onto the stretched grid: (a) altitude, (b) electron density, (c) magnetic-field magnitude, and (d) magnetic-field angle relative to the local vertical.

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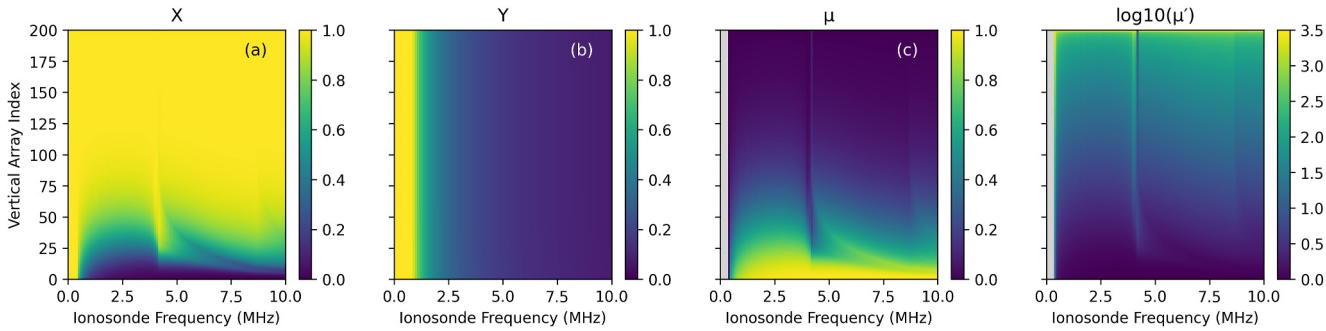


Figure 4. Intermediate quantities computed on the stretched grid: (a) X, (b) Y, (c) O-mode phase refractive index μ, and (d) O-mode group refractive index μ'.

In `PyRayHF`, the group refractive index μ' is computed analytically using Equation 2. The phase refractive index μ depends explicitly on X and Y in the Appleton–Hartree formulation irrespective of propagation mode. Using the chain rule, the derivative $\partial\mu/\partial f$ may therefore be computed as

$$\frac{\partial \mu}{\partial f} = \frac{\partial \mu}{\partial X}\frac{\partial X}{\partial f} + \frac{\partial \mu}{\partial Y}\frac{\partial Y}{\partial f}, \tag{14}$$

which results in the following expression when the frequency derivatives of X and Y are evaluated:

$$\mu' = \mu - \left(2X\,\frac{\partial \mu}{\partial X} + Y\,\frac{\partial \mu}{\partial Y}\right). \tag{15}$$

The appropriate form of μ is substituted into Equation 15 for the given mode and the expression is evaluated to obtain μ'. This analytic approach avoids finite-difference approximations and improves numerical stability near the reflection region, where μ' varies most rapidly.

Panels (c) and (d) of Figure 4 show the resulting phase and group refractive indices, μ and μ'. The phase refractive index decreases from unity near the ground to zero at the reflection height. The group refractive index behaves inversely and increases sharply near the cutoff. The logarithmic color scale in panel (d) highlights the steep rise of μ' in the reflection region, where the stretched grid ensures accurate numerical integration.

To compute the virtual height, a matrix of size $[N_f, N_g]$ is formed containing the distances between adjacent grid points. This matrix is multiplied element-wise by the corresponding μ' matrix and summed along the vertical dimension, yielding the virtual height from Equation 11. Figure 5 shows the resulting virtual-height traces for the O- and X-modes, shown as red and blue curves, respectively. As expected, the X-mode trace is shifted to slightly higher frequencies than the O-mode trace because its reflection condition differs from the O-mode cutoff.

Under the conditions shown, computing a single virtual-height trace for 88 ionosonde frequencies requires only 0.004 s on a typical desktop workstation. This performance makes the vertical forward operator well suited for iterative minimization routines, where the forward model must be evaluated repeatedly to fit observed ionograms or optimize ionospheric parameters.

4. Two-Dimensional Forward Operators

The vertical forward operator described in Section 3 provides an efficient solution for vertically propagating ionosonde signals. However, many HF systems transmit and receive signals at elevation angles less than 90°. Examples include oblique ionosondes, the Super Dual Auroral Radar Network (SuperDARN) HF radars (Chisham et al., 2007; Greenwald et al., 1995), and other ground-based HF diagnostic tools. In such cases, a one-dimensional formulation is insufficient because the wave path bends horizontally as it refracts through the ionosphere.

To represent oblique propagation, `PyRayHF` implements a suite of four two-dimensional forward operators. Two of the solvers assume a uniform, stratified ionosphere and propagate HF rays using Snell's law. These

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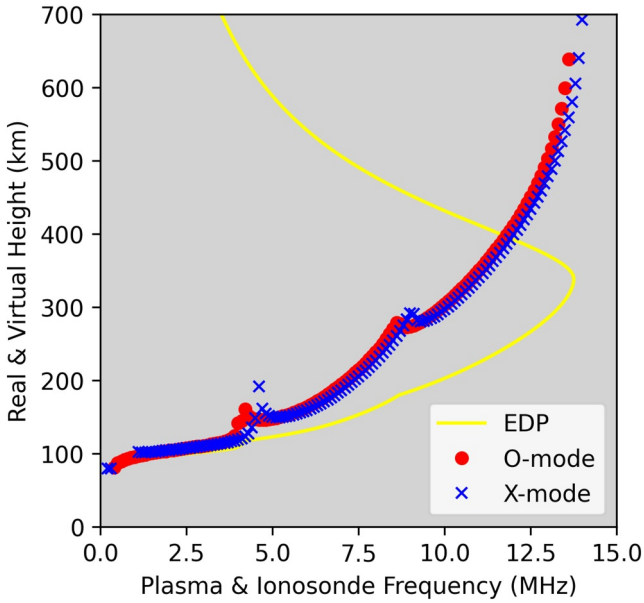


Figure 5. Calculated virtual heights for the O- and X-modes shown relative to the electron density profile (EDP, yellow curve) from Figure 1.

implementations differ in their selected coordinate systems. One assumes a flat Cartesian grid while the other accounts for the spherical geometry of the Earth and ionosphere. The remaining two solvers compute ray paths by integrating the full two-dimensional eikonal equations through a provided magneto-ionic plasma environment. Once again, these formulations differ in their use of either Cartesian or spherical coordinates. Unlike the Snell's law approach, this method allows for horizontal gradients in the plasma density and is generally more accurate. Together, these solvers form a hierarchy of increasing physical fidelity and computational cost, providing a flexible framework for various application needs.

For all formulations considered here, propagation is restricted to the two-dimensional vertical plane connecting the transmitter and receiver. The corresponding azimuth is therefore fixed by the transmitter-receiver geometry and does not appear explicitly in the governing equations.

Figure 6 compares the four two-dimensional ray-tracing formulations implemented in `PyRayHF` for a 14.5 MHz wave launched at a 30° elevation angle. The background color represents the electron density field, with white and black curves denoting the O- and X-mode ray trajectories, respectively. In panels (a) and (b), the background electron density was obtained from the PyIRI model at a single geographic location at 4.5°N and 150°W for 1 September 2025, 00:00 UT. In panels (c) and (d), a horizontal density gradient was imposed by reducing the electron density by up to 50% across the computational domain, with the maximum reduction occurring at the far edge of the grid relative to the transmitter. In all panels, the light-blue vectors indicate the local magnetic field orientation.

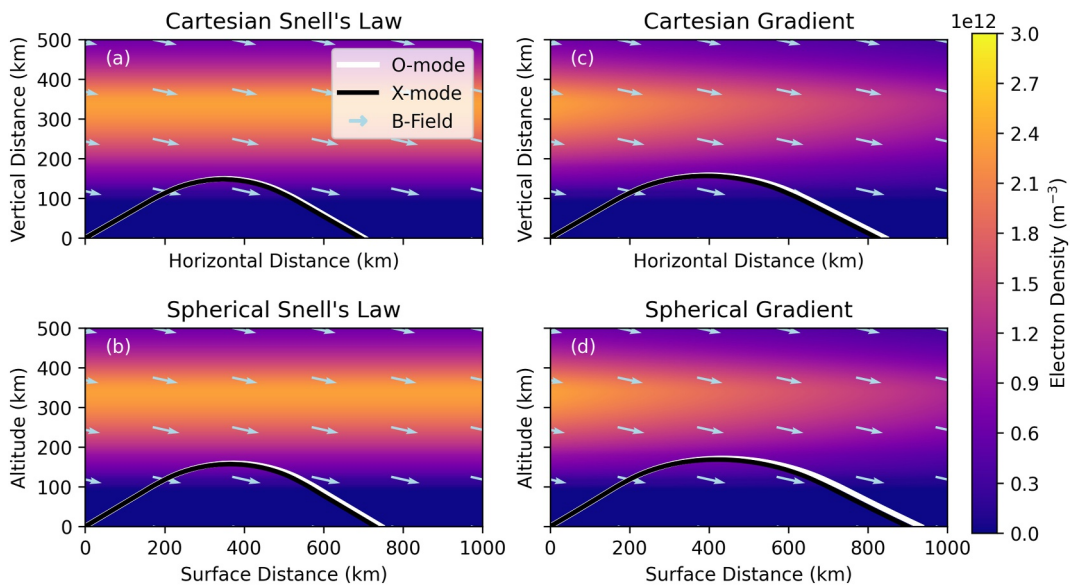


Figure 6. Comparison of the four two-dimensional ray-tracing formulations implemented in `PyRayHF`. Each panel shows the resulting path of a 14.5 MHz wave launched at a 30° elevation angle. The background color represents the electron density field, with white and black curves denoting the O- and X-mode ray trajectories, respectively. Light-blue vectors indicate the local magnetic field orientation. (a) Cartesian ray tracing using Snell's law in a horizontally stratified ionosphere. (b) Spherical ray tracing using Snell's law in (r, ψ) coordinates, conserving $r\,\mu(r)\sin\phi(r)$ along the propagation path. (c) Cartesian ray tracing using the gradient formulation. (d) Spherical ray tracing using the gradient formulation. In panels (a) and (b), the background electron density was obtained from the PyIRI model at a single geographic location at 4.5°N and 150°W for 1 September 2025, 00:00 UT. In panels (c) and (d), a horizontal density gradient was imposed by reducing the electron density by up to 50% across the computational domain, with the maximum reduction occurring at the far edge of the grid relative to the transmitter.

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4.1. Cartesian Ray Tracing Using Snell's Law

The simplest two-dimensional formulation traces ray paths in Cartesian coordinates using Snell's law. This approach assumes a horizontally stratified ionosphere, where input parameters vary only with altitude. Under this approximation, the horizontal component of the wave vector is conserved, leading to the Snell invariant

$$\mu(z)\sin\phi(z) = \mu_0 \sin\phi_0 = \text{constant}, \tag{16}$$

where $\mu(z)$ is the phase refractive index from the Appleton–Hartree equation, $\phi(z)$ is the local propagation angle relative to the vertical, and ϕ_0 is the launch angle.

As altitude increases and the refractive index decreases, the propagation angle increases to preserve the invariant. When μ approaches zero, the ray reaches a turning point and begins to descend, producing the familiar oblique reflection geometry observed in HF propagation. The ray trajectory is obtained by integrating

$$\Delta x_i = \tan\phi_i \, \Delta z, \qquad \Delta s_i = \frac{\Delta z}{\cos\phi_i}, \tag{17}$$

where Δx_i is the horizontal displacement and Δs_i is the incremental path length.

Reflection is detected directly from the refractive index when μ becomes zero or non-real, rather than by enforcing analytic cutoff conditions. After reaching the turning point, the upward trajectory is mirrored to construct the full ray path. This formulation is efficient, requiring approximately 0.0018 s per ray, and is suitable for use in iterative inversion and DA workflows.

4.2. Spherical Ray Tracing Using Snell's Law

The Cartesian Snell's-law formulation neglects Earth's curvature and is therefore limited to short or moderately oblique paths. For long-range propagation, the curvature of both the Earth and ionosphere must be included. `PyRayHF` implements a spherical extension of Snell's law formulated in polar coordinates.

Rays are traced in the (r, ψ) plane, where r is the geocentric radius and ψ is the central angle along Earth's surface. The ionosphere is represented as concentric shells, with plasma parameters depending only on altitude above the Earth radius R_E. The Snell invariant generalizes to

$$r\,\mu(r)\sin\phi(r) = R_E\,\mu_0 \sin\phi_0 = \text{constant}, \tag{18}$$

which reduces to the flat-Earth form in the limit $R_E \to \infty$.

Ray propagation is obtained by integrating

$$\Delta\psi_i = \frac{\tan\phi_i}{r_i}\Delta r_i, \qquad \Delta s_i = \frac{\Delta r_i}{\cos\phi_i}, \tag{19}$$

with reflection occurring when μ becomes zero or imaginary. Compared with the Cartesian case, curvature shifts the reflection point outward and modifies the apparent ground range relative to the flat-Earth solution, as seen in Figure 6b. A single ray path requires approximately 0.0024 s to compute, retaining high efficiency while improving accuracy for long-range propagation.

4.3. Cartesian Ray Tracing Using the Gradient Method

The Snell's-law solvers assume horizontal stratification and therefore cannot trace through horizontal gradients, tilts, or structured density variations. To model such conditions, `PyRayHF` includes a Cartesian gradient-based ray tracer that numerically integrates the full eikonal equations using the explicit Runge-Kutta-Fehlberg (RK45) numerical scheme. These equations provide a generalized representation of ray paths through a medium and are defined as follows:

$$\frac{d\mathbf{r}}{ds} = \mathbf{v}, \qquad \frac{d\mathbf{v}}{ds} = \frac{1}{\mu}(\nabla\mu - (\nabla\mu \cdot \mathbf{v})\,\mathbf{v}), \tag{20}$$

where $\mathbf{r} = (x, z)$ is the ray position and $\mathbf{v}$ is the unit tangent vector.

The refractive-index gradient $\nabla\mu$ is computed directly from the modeled electron density and magnetic-field geometry. Group delay and path length are accumulated using the group refractive index μ'. This formulation accommodates arbitrary two-dimensional refractive-index structure but is substantially more computationally expensive than the Snell's-law solvers.

To introduce a controlled horizontal density gradient, the background electron density field used in the Snell's-law examples was modified to decrease linearly with horizontal distance. Figure 6c shows the resulting Cartesian gradient ray trajectories. The differences between Figures 6a and 6c arise solely from the imposed horizontal density gradient.

A single ray trajectory requires approximately 0.36 s to compute using this method, roughly two orders of magnitude slower than the Snell's-law formulations. Despite this cost, the Cartesian gradient method provides a general two-dimensional forward model that is well suited for validation and sensitivity studies.

4.4. Spherical Ray Tracing Using the Gradient Method

The Cartesian gradient formulation assumes a flat-Earth geometry and therefore becomes inaccurate for long-range propagation. The spherical gradient solver extends the eikonal equations to polar coordinates, ensuring geometric consistency for global-scale ray paths.

Rays are traced in the (r, ψ) plane, with refractive-index gradients containing both radial and tangential components modified by curvature terms. As a ray ascends, its local tangent diverges with increasing radius, lengthening the propagation path and altering the apparent curvature relative to the flat-Earth case. Propagation terminates when the phase refractive index becomes zero or non-real.

Figure 6d illustrates spherical gradient ray tracing under the same horizontally graded background shown in Figure 6c. In this case, the inclusion of Earth's curvature results in a longer apparent propagation path compared to the Cartesian gradient solution. This behavior is consistent with the differences observed between the spherical and Cartesian Snell's-law solutions in Figures 6b and 6a, where curvature similarly increases the total path length.

This formulation provides the most physically complete two-dimensional forward model in `PyRayHF` but at increased computational cost. Each integration step requires refractive-index gradient evaluation in curved coordinates, trigonometric scaling, and frequent tangent-vector renormalization. Adaptive step-size control further increases runtime near turning points. A single ray trajectory typically requires approximately 2.45 s, an order of magnitude slower than the Cartesian gradient solver, while remaining numerically equivalent in accuracy.

5. Verification of Two-Dimensional Forward Operators

To verify the internal consistency of the `PyRayHF` two-dimensional forward operators, all four ray-tracing solvers were evaluated under identical conditions corresponding to a horizontally stratified ionosphere and a flat-Earth geometry, formally defined by the limit $R_E \to \infty$. Under these conditions, the refractive index depends only on altitude, and both geometric curvature and horizontal gradients are eliminated. As a result, the Cartesian Snell's-law, Spherical Snell's-law, Cartesian Gradient, and Spherical Gradient formulations are expected to produce identical ray trajectories.

Figure 7 shows the resulting paths for a 14.5 MHz wave launched at a 30° elevation angle propagating in X-mode. The four solutions closely coincide, confirming that each implementation reproduces the same physical behavior in the stratified limit. The overlaid curves correspond to, in increasing order of thickness, the Cartesian Snell's-law (blue), Cartesian Gradient (green), Spherical Snell's-law (orange), and Spherical Gradient (red) formulations. The near-complete overlap of the trajectories demonstrates that the gradient-based and Snell's-law solvers produce highly consistent ray paths when curvature and horizontal gradients are neglected. Minor deviations in ground range (<10 km) are visible in the presented case due to accumulated numerical differences between the various implementations; nevertheless, the overall agreement remains excellent.

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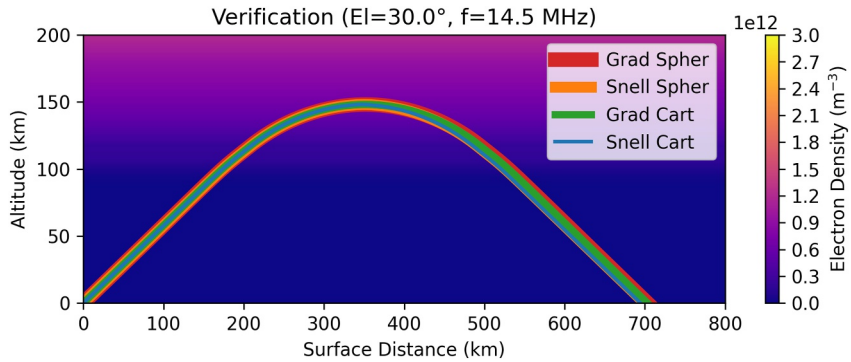


Figure 7. Verification of the four two-dimensional forward operators under horizontally stratified, flat-Earth conditions ($R_E \to \infty$). The red curve represents the spherical gradient solution, the orange curve represents the spherical Snell's-law solution, the green curve represents the Cartesian gradient solution, and the blue curve represents the Cartesian Snell's-law solution. The four trajectories closely overlap, confirming relative consistency among the solvers in the stratified limit.

6. Validation Against a Full 3D Magneto-Ionic Ray Tracer

In the previous section, the `PyRayHF` ray tracing methods were verified against one another under horizontally stratified, flat-Earth conditions. While such internal consistency tests are necessary, they do not demonstrate physical accuracy under realistic ionospheric conditions. It is therefore essential to evaluate the `PyRayHF` solvers against a well-established ray-tracing model that does not rely on the geometric or stratification assumptions employed here. For this purpose, a full 3D magneto-ionic Jones-Stephenson HF ray tracer (Jones & Stephenson, 1975b; Zawdie et al., 2016) was used to produce a reference solution that `PyRayHF` outputs were then validated against.

The validation experiment proceeds as follows: A global ionospheric specification was generated using PyIRI v0.1.5 (V. V. Forsythe, Servan-Schreiber, et al., 2025) for 1 September 2025 at 00:00 UT. Figure 8a shows the resulting global distribution of $foF2$, with IGRF magnetic-field strength, also obtained through PyIRI, overlaid as contours. Transmitters are placed on a 10° × 20° geographic grid. For clarity, only a subset of transmitter locations is shown as white circles, with northward look directions indicated by white lines.

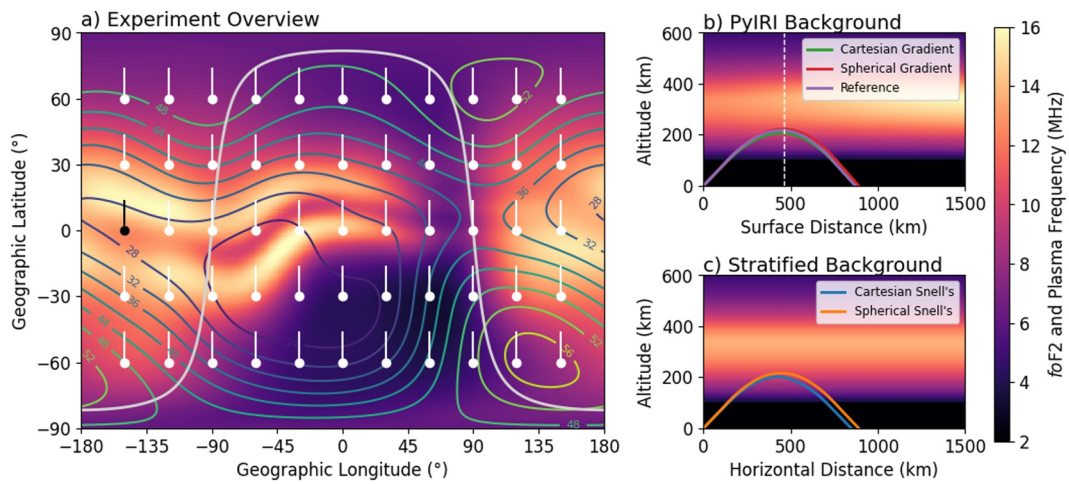


Figure 8. (a) Validation experiment setup. The background map shows $foF2$ from PyIRI for 1 September 2025, 00:00 UT, with magnetic-field strength overlaid as contours. The gray curve indicates the solar terminator. White circles denote a reduced set of transmitter locations, and white lines indicate their northward look directions. (b) Ray-tracing results through the full PyIRI background for the transmitter marked by the black circle in panel (a). The background color shows the local plasma frequency using the same color scale as panel (a). The rays correspond to the Cartesian Gradient (green), Spherical Gradient (red), and reference (purple) solvers. The white dashed curve marks the apex of the Spherical Gradient ray. (c) Cartesian Snell's (blue) and Spherical Snell's (orange) ray paths computed through a stratified electron-density profile extracted at the apex location identified in panel (b).

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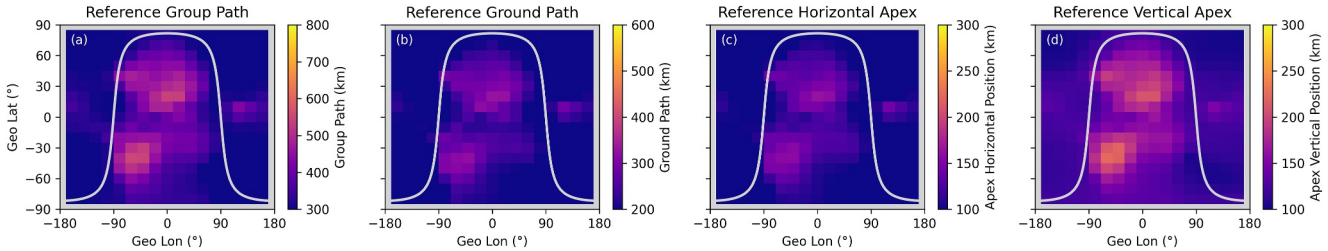


Figure 9. Reference solution used for PyRayHF validation for the 45° elevation and 75% foF2 case. Panels show (a) group path, (b) ground path, (c) horizontal apex location, and (d) vertical apex location. The gray curve indicates the location of the solar terminator.

At each transmitter location, the reference ray tracing model was executed for launch elevation angles of 35°, 45°, and 55°. For each elevation, ray frequencies were set to 50%, 75%, and 90% of the foF2 value at the location of the transmitter. Figures 8b and 8c present a representative two-dimensional slice for the transmitter marked by the black circle in panel (a), located at 150°W and 0°N. The coordinate origin (0, 0) corresponds to the transmitter, and the horizontal axis is aligned with the northward look direction.

Panel (b) shows ray trajectories for the full two-dimensional PyIRI background at 45° elevation and 75% of foF2. The Cartesian Gradient (green), Spherical Gradient (red), and reference (purple) solutions are shown. Because the Snell's-law solvers require a one-dimensional, stratified medium, their rays are computed through an electron-density profile extracted at the apex of the Spherical Gradient solution, indicated by the white dashed curve. The corresponding Cartesian (blue) and Spherical (orange) Snell's-law trajectories are shown in panel (c).

Figure 9 shows the reference solution group path, ground path, horizontal apex location, and vertical apex location for the 45° elevation and 75% foF2 case. In each panel, the gray curve marks the location of the solar terminator. All four parameters exhibit similar global spatial distributions, with larger values on the dayside and near the equator and smaller values on the nightside, reflecting the strong coupling among these ray characteristics.

For each of the nine combinations of elevation angle and frequency, percent differences between the reference solution and the four PyRayHF solvers were computed for the X-mode group path, ground distance, and apex locations. Figure 10 shows the global distributions of percent errors in group path for the 45° elevation and 75% foF2 case, along with the corresponding histograms. As shown, the spatial error patterns are broadly similar across all solvers. Elevated errors occur near the day-night boundary and in the vicinity of the geomagnetic poles. The histogram panels indicate that group path outputs generally lie within 10% of the reference solution for all methods, with a small number of notable outliers discussed below.

Figure 11 highlights two cases where PyRayHF exhibits large discrepancies from the reference solution. Figure 11a depicts a case where the group path is overestimated, located at 80°N and 160°E. This location corresponds to the top-right grid cell in the top row of Figure 10. In this case, the F2 layer exhibits a very thin bottomside and is centered slightly above 200 km, producing an unusually sharp vertical refractive-index gradient. Under these conditions, ray trajectories lie near the transition between reflection and guided propagation, where the governing equations become highly sensitive to small numerical and geometric differences. As a result, modest variations in integration strategy, cutoff detection, or interpolation lead to substantial differences in ground range among the PyRayHF solvers and the reference solution. Specifically, the Cartesian Snell's and Spherical Gradient rays nearly enter Pedersen-type propagation and travel a significant horizontal distance along the ionospheric layer, while the remaining PyRayHF methods fully refract earlier.

Figure 11b demonstrates a case where the group path is underestimated, located at 30°S and 80°E. This location corresponds to the most negative grid cell in the top row of Figure 10. In this case, all PyRayHF solvers agree closely with one another, indicating consistent numerical behavior. However, the full-physics 3D ray tracer signal penetrates deeper before fully refracting, thereby achieving a greater ground range than the PyRayHF solutions.

Figure 12 summarizes the percent error distributions averaged over all nine test cases for four output parameters. Panels show errors in X-mode (a) group path, (b) ground path, (c) horizontal apex location, and (d) vertical apex location. Blue, orange, green, and red denote the four PyRayHF solvers. Table 1 provides the corresponding

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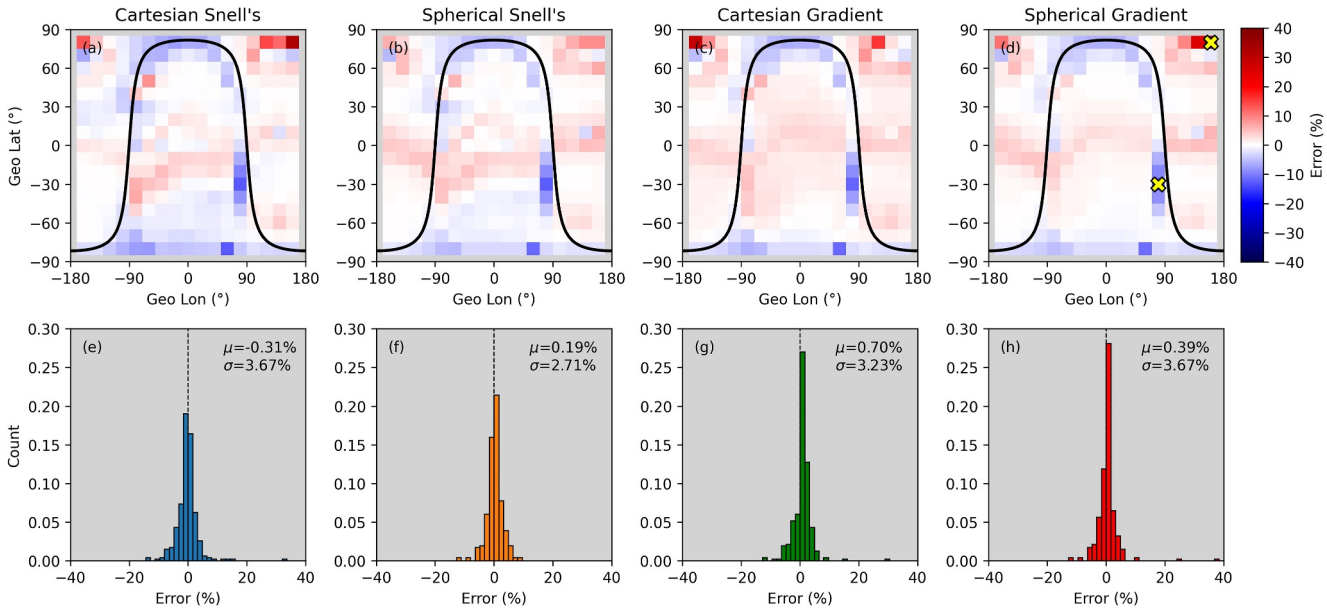


Figure 10. Global distribution of percent errors in group path for (a) Cartesian Snell's, (b) Spherical Snell's, (c) Cartesian Gradient, and (d) Spherical Gradient methods relative to the reference solution. The black curve indicates the solar terminator. Yellow "X" symbols mark the outliers represented by the ray paths in Figure 11. Panels (e)–(h) show the corresponding probability density histograms of percent errors. Histogram colors follow the same color scheme as in Figure 8. The mean μ and standard deviation σ are shown for each histogram.

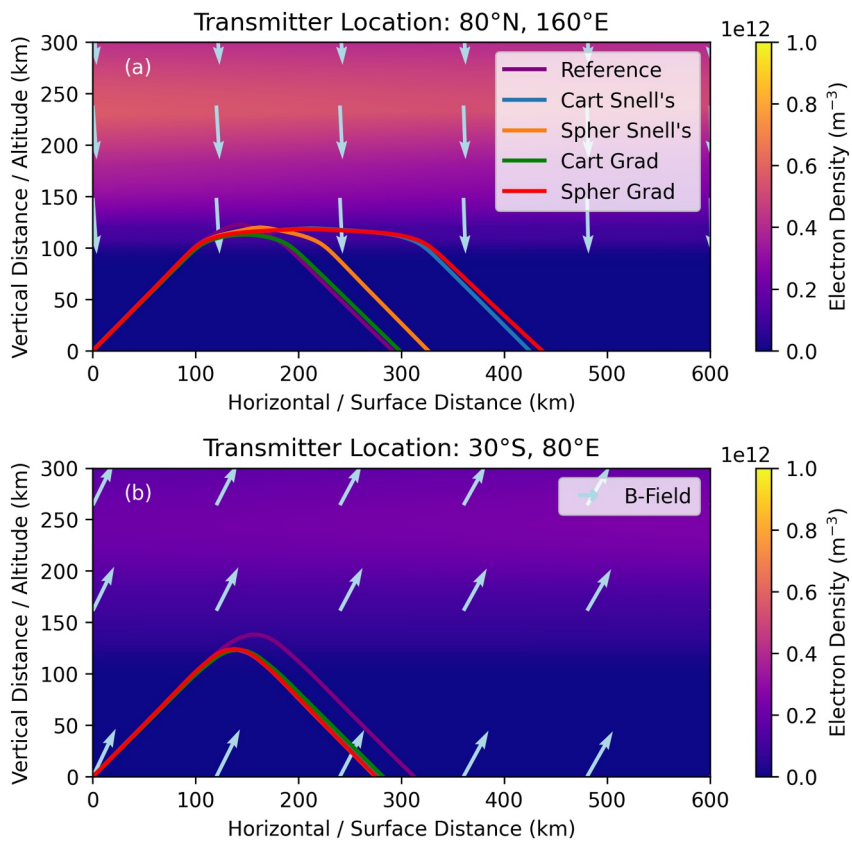


Figure 11. Comparison between the reference solution and `PyRayHF` ray tracers at (a) 80°N, 160°E and (b) 30°S, 80°E. The rays correspond to Cartesian Snell's (blue), Spherical Snell's (orange), Cartesian Gradient (green), Spherical Gradient (red), and reference (purple) solvers.

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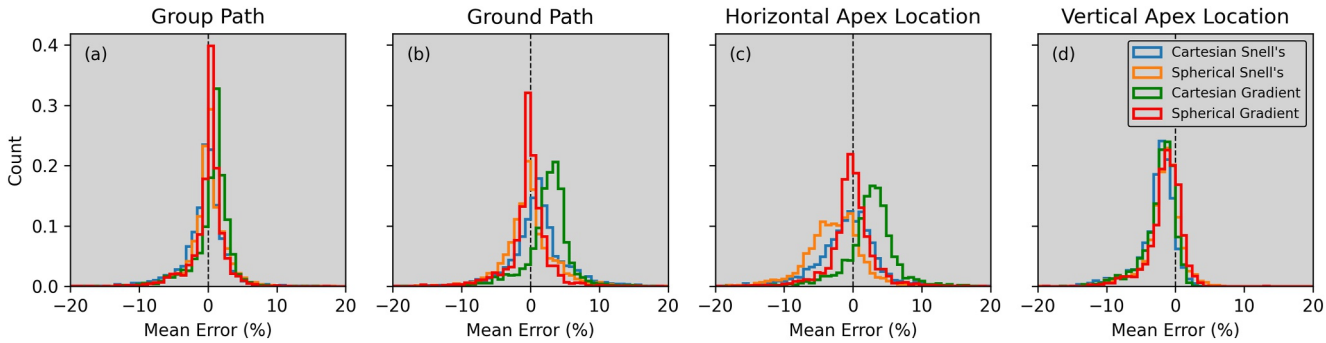


Figure 12. Probability density functions of percent error from the reference solution for Cartesian Snell's (blue), Spherical Snell's (orange), Cartesian Gradient (green), and Spherical Gradient (red) methods for (a) group path, (b) ground path, (c) horizontal apex location, and (d) vertical apex location.

mean and standard deviation statistics for all depicted distributions. Each method produces group path errors centered near zero and generally confined to within 10%. Larger differences are observed for ground path and horizontal apex location, where geometric assumptions exert greater influence. The Spherical Gradient solver shows the best overall agreement with the reference solution, reflecting its consistent treatment of horizontal gradients. The Cartesian Gradient solver exhibits a systematic positive bias of approximately 3% in ground path and horizontal apex location due to its flat-Earth geometry. The Spherical Snell's solver displays a systematic negative bias of approximately 3% in apex location, indicating underestimation associated with its stratified-medium assumption.

All four ray-tracing methods produce nearly identical results for the vertical location of the ray apex. For this parameter, percent errors remain within approximately 10%, with a small and consistent negative bias across all solvers. This agreement reflects the dominant role of the vertical refractive structure in controlling apex height, which is treated similarly by all four formulations.

Among the evaluated outputs, the group path is the most important observable for oblique ionogram interpretation. Across all nine test cases, all four `PyRayHF` solvers demonstrate strong agreement with the reference solution group path. Notably, even the simplest Cartesian Snell's-law implementation achieves accuracy comparable to the more sophisticated solvers. This result indicates that increased geometric or numerical complexity does not necessarily yield substantial improvement for group path estimation. Accordingly, for data-assimilation applications in which repeated forward evaluations are required, the choice of ray-tracing formulation should be guided primarily by computational efficiency. The following section therefore examines the computational cost and scalability of each method.

7. Computational Considerations and the Need for Simplified Operators

The previous section demonstrated that all four `PyRayHF` ray tracers reproduce the full 3D magneto-ionic ray tracer results to within approximately 15% across all evaluated metrics, with the Spherical Gradient solver providing the best overall agreement. While this demonstrates the physical validity of the solvers, their applicability to data-assimilation workflows depends critically on computational feasibility.

Table 1
Mean and Standard Deviation of the Percent Error Distributions of Each Ray Parameter Computed for All Four `PyRayHF` *Ray-Tracing Methods*

Method	Group path	Ground path	Horizontal apex	Vertical apex
Cartesian Snell's	−0.57% ± 3.32%	1.03% ± 4.51%	−0.01% ± 4.72%	−2.67% ± 2.73%
Spherical Snell's	−0.06% ± 3.02%	−0.27% ± 4.04%	−2.86% ± 4.37%	−1.81% ± 2.69%
Cartesian Gradient	0.53% ± 3.19%	2.45% ± 3.92%	2.49% ± 4.15%	−2.38% ± 2.53%
Spherical Gradient	0.00% ± 3.03%	−0.32% ± 3.66%	−0.25% ± 3.80%	−1.53% ± 2.45%

Note. Values are reported as mean ± standard deviation and correspond to the histograms in Figure 12.

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Table 2
Estimated Total Wall Times for Repeated Forward Evaluations Assuming $N_{elev} = 20$, $N_{iter} = 50$, *and* $N_f = 20$

Forward operator	t_{ray} (s)	T_{tot} (s)	T_{tot} (min)
Cartesian Snell's Law	1.78×10^{-3}	35.6	0.59
Spherical Snell's Law	2.41×10^{-3}	48.2	0.80
Cartesian Gradient	3.6×10^{-1}	3.6×10^{4}	600.0
Spherical Gradient	2.45	2.45×10^{5}	4083.0

Note. The runtimes t_{ray} correspond to single-ray computations.

In DA, each update of the background state requires repeated forward-model evaluations within an iterative minimization procedure. For oblique ionogram inversion, these evaluations occur across three nested loops. First, multiple launch elevation angles must be scanned to identify rays that reach the receiver. Second, this search is repeated for every iteration of the minimization. Third, the entire procedure is carried out independently for each sounding frequency. If N_{elev} denotes the number of launch angles, N_{iter} the number of minimization iterations, and N_f the number of transmitted frequencies, the total number of ray computations per DA update is

$$N_{tot} = N_{elev} \times N_{iter} \times N_f. \tag{21}$$

A representative configuration requires approximately $N_{elev} = 20$ elevation angles to home a ray onto the receiver, $N_{iter} = 50$ iterations in a typical minimization loop, and $N_f = 20$ frequencies in a single oblique ionogram. This results in roughly 2×10^4 ray integrations per update of the background state.

Table 2 summarizes the estimated wall-clock times for a desktop workstation under these conditions. Even the efficient Snell's-law solvers become nontrivial when executed tens of thousands of times per assimilation window, and they remain restricted to horizontally stratified ionospheres. The gradient-based solvers remove this restriction but are computationally prohibitive for practical DA applications.

Even under this modest configuration, the computational gap between the Snell's-law and gradient-based solvers spans nearly four orders of magnitude. This disparity makes it impractical to embed full two-dimensional or spherical gradient ray tracing within DA cycles that require thousands of forward evaluations per assimilation window.

A faster approximation is therefore required. Such an approach must preserve the dominant physics governing group delay while avoiding explicit integration of the full oblique ray path. The method adopted here combines the vertical forward operator with a midpoint transformation, which replaces the complete oblique trajectory by its vertically equivalent representation through a stratified ionosphere.

8. Midpoint Approximation for Oblique-to-Vertical Conversion

To avoid the computational cost associated with repeated two-dimensional ray tracing within data-assimilation workflows, an approximate geometric mapping is employed to convert an oblique ionogram into its vertically equivalent representation. This conversion assumes a curved Earth, a flat ionosphere, and follows the classical midpoint relations derived from Martyn's theorem (Martyn, 1935), which are widely used in ionogram interpretation (Budden, 1985; Davies, 1990) and reproduced here.

The midpoint transformation implemented in `PyRayHF`'s oblique_to_vertical routine provides estimates of the virtual reflection height and equivalent vertical frequency at the midpoint between the transmitter and receiver. These quantities are obtained without explicit integration of the full oblique ray path, thereby significantly reducing computational cost.

Let D denote the ground range between the transmitter and receiver and p the total group path. The central angle subtended by the transmitter and the midpoint at the Earth's center is

$$\theta = \frac{D}{2R_E}, \tag{22}$$

and the corresponding correction for Earth curvature is

$$\Delta h_{curv} = R_E(1 - \cos\theta). \tag{23}$$

The local incidence angle at the midpoint follows directly from the oblique geometry,

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$$\phi = \arcsin\left(\frac{D}{p}\right). \tag{24}$$

Using the midpoint approximation (Budden, 1985; Davies, 1990; Martyn, 1935), the virtual reflection height is computed as

$$h_v = \frac{1}{2} p \cos\phi - \Delta h_{curv}, \tag{25}$$

and the equivalent vertical frequency is obtained by scaling the transmission frequency f of the oblique ray

$$f_v = f \cos\phi. \tag{26}$$

The resulting pair (f_v, h_v) defines the virtual height trace that would be observed for a vertically incident wave propagating through a stratified ionosphere equivalent to that sampled by the oblique path. This midpoint conversion preserves the dominant magneto-ionic dispersion effects while avoiding the computational overhead of full oblique ray tracing.

The accuracy of the midpoint approximation depends primarily on the degree to which the ionosphere can be approximated as horizontally stratified and on how representative the midpoint geometry is of the refractive structure encountered along the full oblique path. Departures from these assumptions, such as strong horizontal gradients or localized ionospheric structures, will inevitably introduce discrepancies relative to full ray-tracing solutions. However, the results presented in the preceding sections demonstrate that the dominant contribution to the observed group path is controlled by the vertically integrated ionospheric structure, which is well captured by the midpoint-based formulation. As a result, this approximation provides a practical balance between physical fidelity and computational cost for data-assimilation applications.

In the following section, minimization strategies within `PyRayHF` are discussed in the context of this midpoint-based forward model, with particular emphasis on its application to the ANCHOR data-assimilation framework.

9. Parameter Minimization Within the ANCHOR Data Assimilation Framework

The expressed objective of `PyRayHF` is to facilitate the incorporation of oblique ionogram measurements into operational DA frameworks. `PyRayHF` was specifically developed to integrate seamlessly with the U.S. Naval Research Laboratory's ANCHOR system. ANCHOR (V. V. Forsythe, McDonald, et al., 2024) is a parameterized data-assimilation framework designed to combine heterogeneous ionospheric observations into self-consistent, time-evolving maps of electron density and related parameters. Rather than assimilating raw measurements directly, ANCHOR extracts a set of representative anchor points from each data source during the data-aggregation stage. These parameters are subsequently merged into two-dimensional forecast fields using a Kalman filter, which updates the model state by balancing observational information against the background error covariance (V. V. Forsythe, McDonald, et al., 2024). During preprocessing, ANCHOR already has access to the forecast electron-density fields and the magnetic-field geometry. In a spun-up assimilation cycle, this information provides reliable initial estimates for local minimization procedures that fit model parameters to ingested oblique data. This substantially reduces the effective search space and allows for efficient assimilation of oblique ionogram measurements.

A processed oblique ionogram consists of a set of transmitted frequencies and their associated group paths between a transmitter-receiver pair of known locations. Figure 13a illustrates the construction of a synthetic oblique trace through a PyIRI electron density field. The transmitter is located at 0° W, 0° N, and the receiver is positioned 1,000 km to the north. The three-dimensional Jones-Stephenson magneto-ionic ray tracer is used to compute homed X-mode rays through this background, and its output is treated as the measured oblique in this analysis. Each ray is color-coded by the plasma density equivalent of its transmitted frequency (Equation 5), using the same color scale as the background field. The apex of each ray is indicated by a small green circle.

Figure 13b shows the synthetic oblique ionogram derived from the ray tracing output in panel (a). The transmission frequency of each ray is plotted against its corresponding group path to form an oblique ionogram. By

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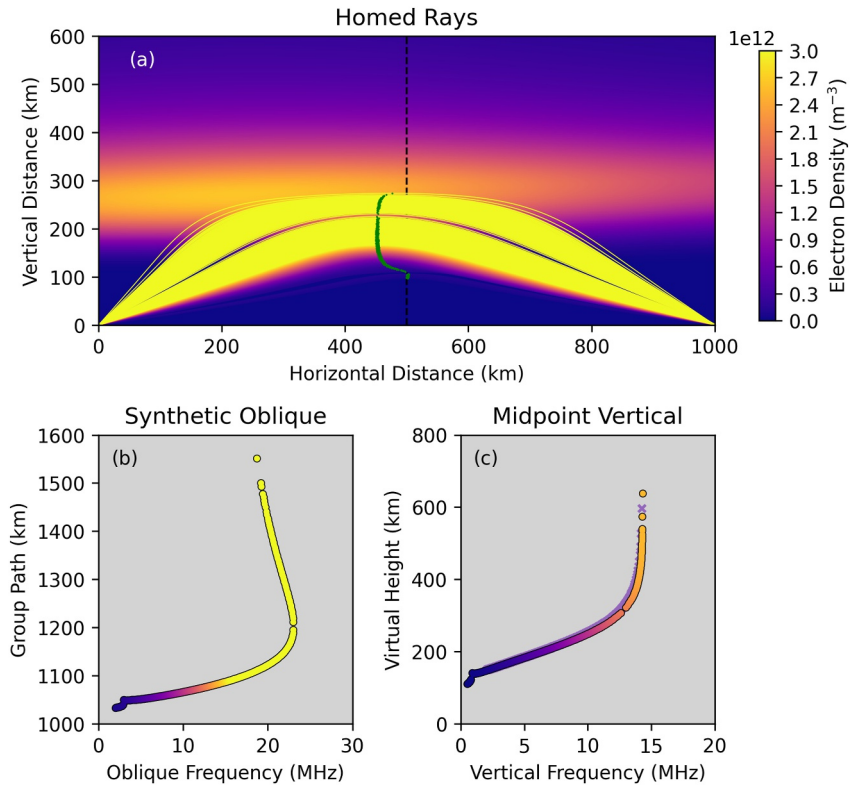


Figure 13. (a) Homed rays between a transmitter and receiver obtained using a 3D magneto-ionic ray tracer for the PyIRI background shown in Figure 8a. The transmitter is located at 0°W, 0°N, and the receiver is 1,000 km to the north. The background color represents electron density, with the corresponding color scale shown on the right. Each ray is colored by the plasma density equivalent to its transmitted frequency, and a small green circle marks the apex of each ray. (b) The corresponding synthetic oblique ionogram, where point color indicates the transmitted oblique frequency. (c) The midpoint-converted vertical ionogram, where point color denotes the equivalent vertical frequency. Purple "X" markers show the synthetic vertical ionogram at the midpoint obtained using the reference ray tracer.

applying the midpoint approximation described in Section 8 (Equations 25 and 26), these oblique data points are converted to an equivalent vertical ionogram shown in Figure 13c. As expected, the critical vertical frequency is lower than the peak frequency of the oblique trace. To verify the midpoint transformation, full 3D magneto-ionic calculations were performed for vertically incident rays at the midpoint location indicated by the dashed line in panel (a). The resulting vertical ionogram, shown as purple "X" markers in Figure 13c, agrees closely with the midpoint-converted values, suggesting that this method of geometric conversion may be sufficient to use when ingesting oblique ionograms into DA systems.

Figure 14 demonstrates the technique of fitting a parametrized electron density profile (EDP) to a midpoint-transformed oblique ionogram. The red line represents the actual EDP taken from the midpoint of Figure 13a (the vertical dashed line). This profile is taken to be the reference truth by which this minimization scheme is evaluated. The dotted red ionogram trace is the same midpoint-transformed oblique ionogram presented in Figure 13c. Within ANCHOR, the forecast fields are sampled at the midpoint location to obtain the local background ionospheric parameters and magnetic-field geometry. These quantities form the background state for the minimization. The nominal background EDP, shown as a blue dotted line, is deliberately perturbed from the truth. Specifically, *Nm*F2 is increased by 50%, *hm*F2 is increased by 20%, and the bottomside thickness parameter *B*0 is reduced by 20%. By applying `PyRayHF`'s vertical forward operator to this background EDP, the blue triangle vertical ionogram is obtained. The fitting algorithm then varies the EDP parameters until the forward-modeled ionogram matches the midpoint-transformed oblique data as closely as possible.

Because the F2 layer dominates HF reflection, it is treated as the primary target of the fit. The peak density *Nm*F2 is derived directly from the maximum observed frequency using the appropriate mode-dependent cutoff

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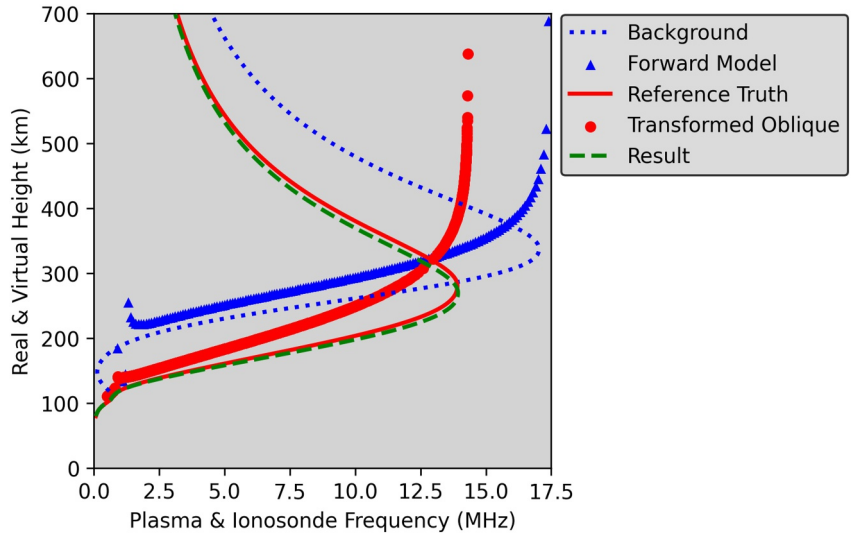


Figure 14. Results of the `PyRayHF` minimization. The red curve shows the EDP taken as the known truth, with red circles marking the midpoint-converted ionogram from Figure 13c. The blue dotted line shows the intentionally perturbed background EDP, and the blue triangles indicate the corresponding `PyRayHF` forward-modeled vertical ionogram. The green dashed curve shows the EDP obtained from the `PyRayHF` minimization.

condition. Two bottomside parameterizations are supported. The first uses a single thickness parameter B_{bot}^{F2}. The second follows the IRI formulation using the parameters $B0$ and $B1$. For the B_{bot}^{F2} formulation, the algorithm varies hmF2 and B_{bot}^{F2} while holding the peak density fixed. Initial estimates of hmF2 and the bottomside parameter are taken from the forecast state, with allowable deviations defined by a user-specified uncertainty, typically 20%. A parameter grid is constructed within these bounds. At each grid point, the vertical forward operator `model_VH` is used to compute synthetic virtual heights. The misfit χ^2 between modeled and observed virtual heights is quantified by summing the square of their difference over every ray frequency f_i in the ionogram data

$$\chi^2 = \sum_i \left[h_{\mathrm{v,model}}\left(f_i\right) - h_{\mathrm{v,obs}}\left(f_i\right)\right]^2. \tag{27}$$

The parameter set minimizing this quantity is selected. The minimization may be performed using either a brute-force grid search or derivative-free methods such as Powell's or Levenberg–Marquardt optimization.

This functionality is implemented in the `minimize_parameters` routine within `PyRayHF`. For each trial parameter set, the routine evaluates the vertical forward model, computes the residual, and returns the parameter combination that best matches the observations. The resulting optimized EDP represents the most probable stratified ionospheric profile consistent with the oblique measurement. Because the search is initialized close to the forecast state, convergence typically requires only a few tens of forward-model evaluations when using derivative-free optimization. This makes the approach practical for large-scale assimilation cycles involving many oblique paths. The results of this fitting algorithm are represented by the green dashed EDP in Figure 14, closely matching the reference data.

To quantify errors introduced by the midpoint approximation, synthetic X-mode oblique ionograms were generated using the reference 3D ray tracer for all transmitter locations shown in Figure 8a. Each oblique ionogram was converted to its vertically equivalent form using the midpoint transformation. The maximum vertical frequency was then identified, and the corresponding critical plasma frequency was computed. Because the true ionospheric state at the midpoint is known, the resulting errors can be directly quantified. Figure 15 shows the absolute and relative foF2 errors introduced by the midpoint approximation. Most cases fall within a 5% error range, with a single outlier near 7%, corresponding to an absolute error of approximately 1 MHz. For a transmitter-receiver separation of 1,000 km, the midpoint approximation therefore introduces errors well below 10%, which is acceptable for data-assimilation applications (V. V. Forsythe, McDonald, et al., 2024).

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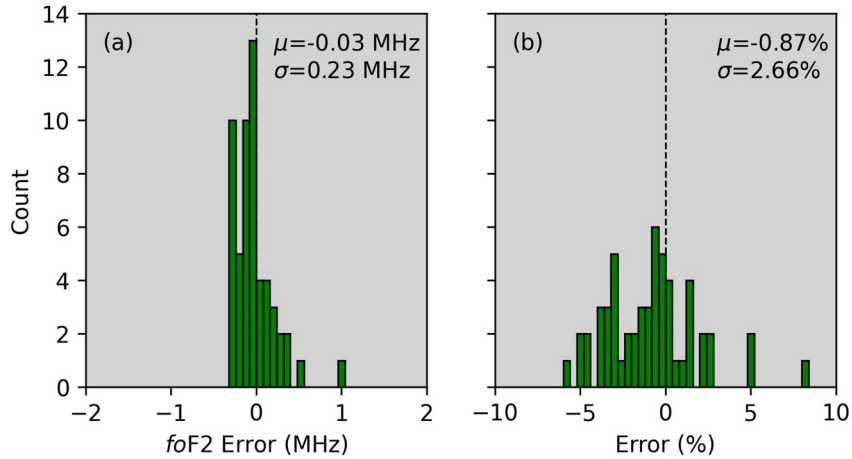


Figure 15. (a) Absolute *fo*F2 errors introduced by the midpoint approximation. (b) The same errors expressed as percentages. The mean μ and standard deviation σ are presented for each distribution.

The purpose of this analysis is not to validate a new ray-tracing model against experimental observations, but rather to evaluate the forward-modeling and parameter-retrieval methodology proposed for oblique-ionogram assimilation. Since the intended application is iterative ionospheric data assimilation, the relevant performance metric is the ability to recover physically meaningful ionospheric parameters from oblique measurements with sufficient accuracy and computational efficiency. A controlled synthetic framework is therefore employed, allowing retrieval errors to be quantified relative to a known reference ionosphere generated using a full three-dimensional magneto-ionic ray tracer. The statistical analyses presented here provide a quantitative assessment of the accuracy and robustness of the proposed methodology and establish its suitability for assimilation applications.

Because the `PyRayHF` minimization derives *Nm*F2 directly from the highest observed frequency rather than varying it explicitly, the final errors in *Nm*F2 remain consistent with those shown in Figure 15, independent of the initial background value. To further quantify errors in *hm*F2 and the bottomside thickness parameter *B*0, the minimization was applied to all synthetic oblique ionograms using different perturbations of the background state. For each parameter, the background value was set to either 20% higher or 20% lower than the truth, resulting in four test cases. Figure 16 summarizes the resulting residual distributions of both background and fit data. Panels (a)–(d) show residuals for *hm*F2, while panels (e)–(h) show residuals for *B*0. In several cases, the minimization successfully reduces the background bias. However, in some cases secondary peaks remain aligned with the background state. This behavior reflects the coupled nature of the *hm*F2 and *B*0 parameters. For example, increasing electron density below the F2 peak can be achieved either by increasing *B*0 or by decreasing *hm*F2. Under such conditions, the minimization may converge to an alternative local minimum that partially preserves the background bias.

The final outputs of the local minimization, including *Nm*F2, *hm*F2, and the selected bottomside shape parameter, are assimilated at the midpoint location as anchor-point measurements. This procedure is fully consistent with the ANCHOR framework (V. V. Forsythe, McDonald, et al., 2024). A detailed description of the subsequent assimilation of anchor-point measurements into the ANCHOR state vector and forecast update is provided in (V. V. Forsythe, McDonald, et al., 2024) and is not repeated here.

10. PyRayHF: Open Research, Tutorials, and Reproducibility

`PyRayHF` is released as an open-source Python package (V. Forsythe, Burrell, Valentine, & Hughes, 2025) that provides a collection of forward operators for high-frequency (HF) ionospheric applications. The package can be installed directly from the Python Package Index (PyPI), which automatically resolves all required dependencies.

To support reproducibility and facilitate adoption, `PyRayHF` includes a set of documented Jupyter Notebook tutorials that demonstrate the workflows used throughout this study. The tutorial Example_Generate_Input_Arrays illustrates how PyIRI is used to generate background ionospheric and magnetic-field inputs for

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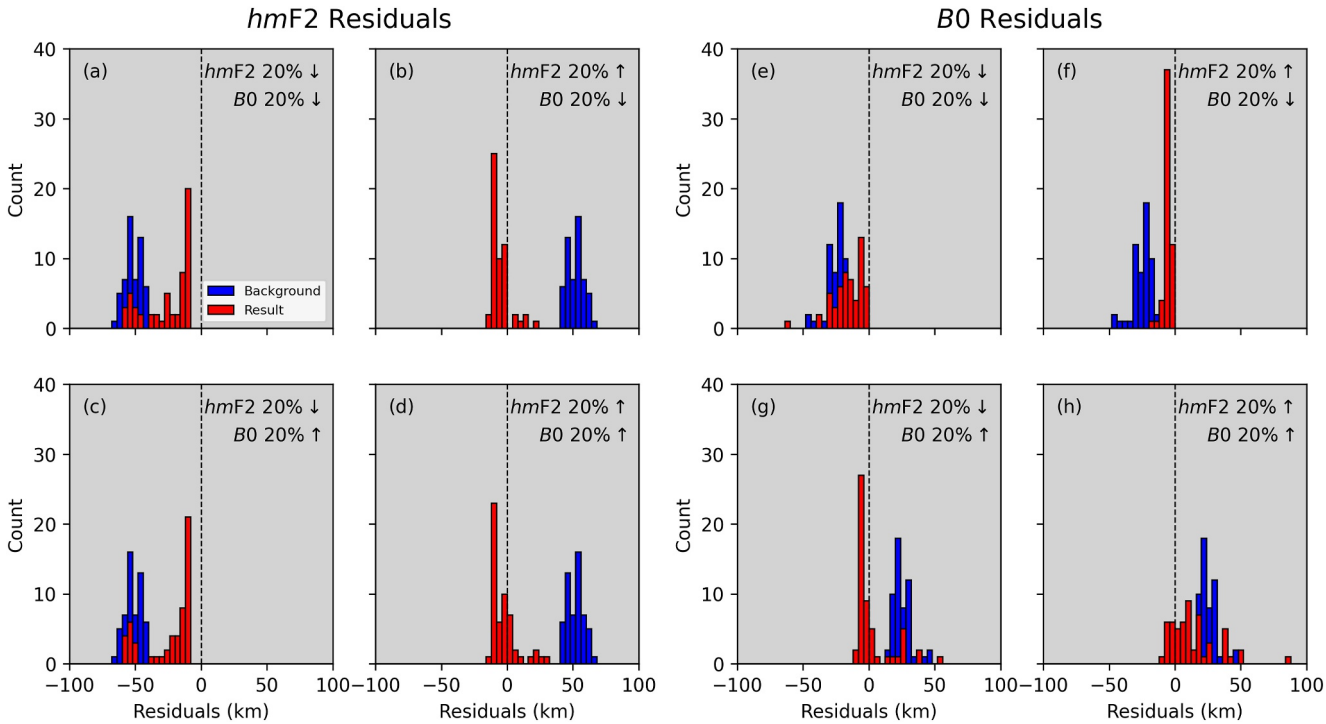


Figure 16. Results of the `PyRayHF` minimization. Panels (a)–(d) show the residual distributions for the *hm*F2 parameter, while panels (e)–(h) show the residuals for the *B*0 parameter. Annotations indicate the deviation of the background parameter from the known truth used to initialize the minimization.

`PyRayHF` and reproduces Figure 1. The tutorial Example_Vertical_Forward_Operator demonstrates the operation of the vertical forward operator and reproduces Figure 2 through Figure 5.

Separate tutorials are provided for each two-dimensional ray-tracing formulation. The tutorials Example_PyRayHF_Cartesian_Snells, Example_PyRayHF_Spherical_Snells, Example_PyRayHF_Cartesian_Gradient, and Example_PyRayHF_Spherical_Gradient reproduce the four ray-tracing panels shown in Figure 6.

The parameter-minimization workflow used for oblique ionogram assimilation is demonstrated in the Example_Minimization tutorial. This example shows how to apply the minimize_parameters routine and reproduces the results shown in Figure 14.

Together, these tutorials provide end-to-end examples that reproduce the key figures and results presented in this paper, enabling transparent validation and straightforward extension of the `PyRayHF` framework.

11. Conclusion

This study introduced `PyRayHF`, a Python-based ray-tracing framework designed to support high-frequency ionospheric modeling and data-assimilation applications. `PyRayHF` implements a hierarchy of forward operators, ranging from magneto-ionic vertical tracing to two-dimensional oblique ray tracing using both Snell's-law and gradient-based formulations in Cartesian and spherical geometries. Together, these operators provide a flexible set of tools for investigating HF propagation physics while enabling systematic evaluation of accuracy and computational cost.

Through verification and validation against a full three-dimensional magneto-ionic ray tracer, it was shown that all `PyRayHF` solvers reproduce key oblique-ray observables with errors generally confined to within 10%–15% of the reference solution. For the primary observable of interest in oblique ionogram interpretation, the group path, even the simplest Cartesian Snell's-law formulation achieves accuracy comparable to more sophisticated gradient-based solvers. These results demonstrate that increased geometric or numerical complexity does not necessarily yield improved performance for the observables most relevant to data assimilation.

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From a data-assimilation perspective, computational efficiency is the dominant constraint. The gradient-based solvers, while physically general, are prohibitively expensive for iterative assimilation workflows, requiring several orders of magnitude more runtime than the stratified Snell's-law formulations. As a result, only stratified two-dimensional approaches are viable for practical DA applications. Given this limitation, explicit two-dimensional ray tracing offers little advantage over a vertically equivalent formulation when the ionosphere can be approximated as horizontally stratified.

To address this trade-off, `PyRayHF` couples a fast vertical forward operator with a midpoint-based oblique-to-vertical transformation. This approach replaces explicit oblique ray tracing with a vertically equivalent representation while preserving the dominant magneto-ionic dispersion effects governing group delay. For transmitter–receiver separations of 1,000 km, the midpoint approximation introduces $foF2$ errors typically below 5%, and remains below 10% even for 2,000 km paths.

Beyond forward modeling, `PyRayHF` provides a complete parameter-minimization framework for oblique ionogram assimilation. The implemented minimization routines directly retrieve *hm*F2 and bottomside shape parameters from midpoint-converted observations, using only a small number of forward-model evaluations. Because this scheme operates entirely within a stratified representation, it is computationally compatible with large-scale, iterative DA systems. The resulting optimized parameters are naturally expressed as anchor-point measurements and can be assimilated directly into parametrized frameworks such as ANCHOR.

In summary, `PyRayHF` demonstrates that efficient, stratified forward operators combined with physically motivated geometric approximations are sufficient for practical assimilation of oblique HF measurements. By avoiding unnecessary geometric complexity, the framework achieves orders-of-magnitude reductions in computational cost while retaining the accuracy required for operational data assimilation. `PyRayHF` thus provides a practical and openly accessible pathway for incorporating oblique HF observations into modern ionospheric DA systems.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The `PyRayHF` software is licensed under MIT and published on GitHub: https://github.com/victoriyaforsythe/PyRayHF.git (V. Forsythe, Burrell, Valentine, & Hughes, 2025). The `PyIRI` software is licensed under MIT and published on GitHub: https://github.com/victoriyaforsythe/PyIRI.git (V. V. Forsythe, Servan-Schreiber, et al., 2025). Figures were generated using `Matplotlib` (version 3.10.3), available under the Matplotlib license at https://matplotlib.org/.

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